



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
19-07-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

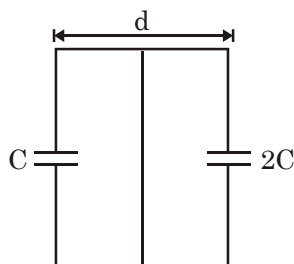
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

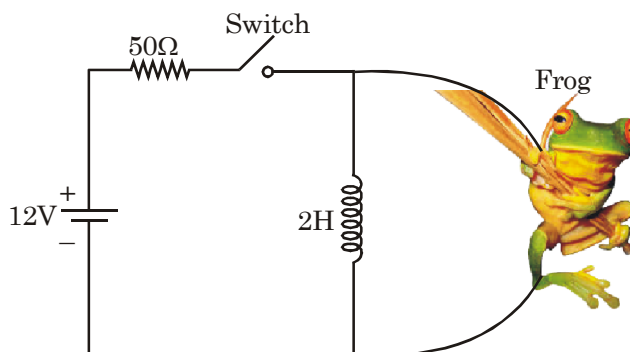
1. Two capacitors are connected by the ideal wires forming a square with side d . The central wire connects the midpoints of the opposite sides of squares as shown. The circuit is placed in the magnetic field, directed into the page, that changes with time as $B(t) = Kt$, where K is positive constant. After sometime the central wire is cut and the magnetic field is then kept constant. The charge on capacitor 'C' after the equilibrium has been reached will be.



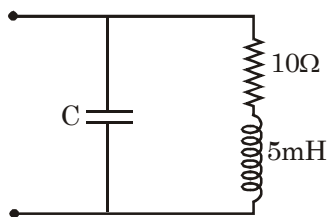
- (A) $\frac{KCd^2}{3}$ (B) $\frac{KCd^2}{6}$ (C) $\frac{KCd^2}{2}$ (D) KCd^2
2. A rope of linear density of 0.2 kg m^{-1} is whirled at a constant angular velocity ω in anticlockwise direction, so that the rope becomes a taut circle of radius R . A kink develops in the whirling rope so that it seems to be moving with a speed of 60 ms^{-1} relative to an observer on the ground in anticlockwise direction. The tension developed in the rope, then must be :-
 (A) 60N (B) 240 N (C) 180 N (D) 120 N

Space for Rough Work

3. The circuit used by biology student to study "Frog Kick". She noticed that frog kicked a little when switch was closed but kicked violently for 5s when the switch was opened. Model the frog as a resistor and calculate its resistance. Assume that it takes 10 mA for the frog to kick violently. [Given $\ln 24 = 3.178$]



- (A) 3.178Ω (B) 0.635Ω (C) 4.852Ω (D) 1.271Ω
4. Two waves $y_1 = a \sin \left(\frac{\pi}{2}x - \omega t \right)$ and $y_2 = a \sin \left(\frac{\pi}{2}x + \omega t + \frac{\pi}{3} \right)$ get superimposed in the region $x \geq 0$. Find the number of nodes in the region $0 \leq x \leq 6\text{m}$
- (A) 4 (B) 6 (C) 3 (D) 5
5. An ideal choke takes a current of 8A when connected to an AC supply of 100 volt and 50 Hz. A pure resistor under same conditions takes 10 Ampere current. Now both of these are connected in parallel to an 150 volt and 40 Hz source. The current through source is :-
- (A) 15 A (B) 5A (C) $15 \sqrt{2}$ (D) 10 A
6. An industrial load is modeled as a series combination of an inductor and resistance as shown in figure. Calculate the value of capacitance C (in μF) across the series combination so that the net impedance is resistive at a angular frequency of 2kHz.



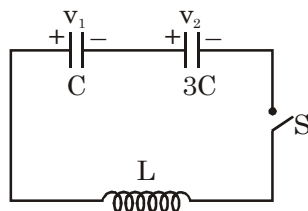
- (A) $5 \mu\text{F}$ (B) $15 \mu\text{F}$ (C) $25 \mu\text{F}$ (D) $50 \mu\text{F}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

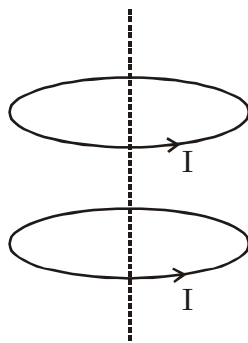
7. The capacitors shown in circuit have capacitance C and $3C$ and they have been charged to potentials $V_1 = 2V_0$ and $V_2 = 3V_0$ respectively. Switch S is closed to connect them to the inductor 'L'. Which of the following is/are correct :-



- (A) Maximum current through the inductor $\frac{5V_0}{2} \sqrt{\frac{3C}{L}}$
- (B) Maximum current through the inductor $\frac{V_0}{2} \sqrt{\frac{3C}{L}}$
- (C) Potential difference across 'C' when the current in inductor is maximum is $\frac{7V_0}{4}$
- (D) Potential difference across '3C' is maximum is $\frac{11V_0}{4}$

Space for Rough Work

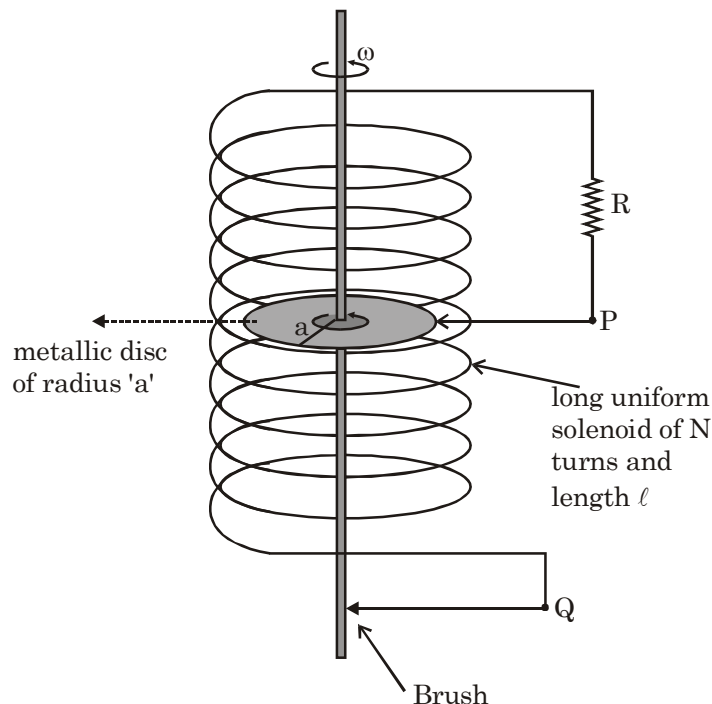
8. Two identical superconducting loops (zero resistance) are positioned co-axially (a line passes through both their centres, perpendicular to each) and far apart. The loop have self inductance 'L'. Equal current 'I' are passing through each of these loops in the same direction. Now bring the loops together which of the following is correct ?



- (A) Magnetic flux passes through each loop increases.
 (B) Magnetic flux passes through each loop remain same
 (C) Current through each loop is $2I$
 (D) Current through each loop is $\frac{I}{2}$

Space for Rough Work

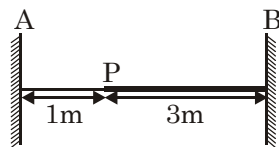
9. A metallic disc of radius 'a' mounted on a rigid axle is rotating with a constant angular velocity ω inside a long solenoid of inductance L whose two ends are connected to the rotating disc by two brush contacts as shown. The total resistance of the whole circuit is R . Then



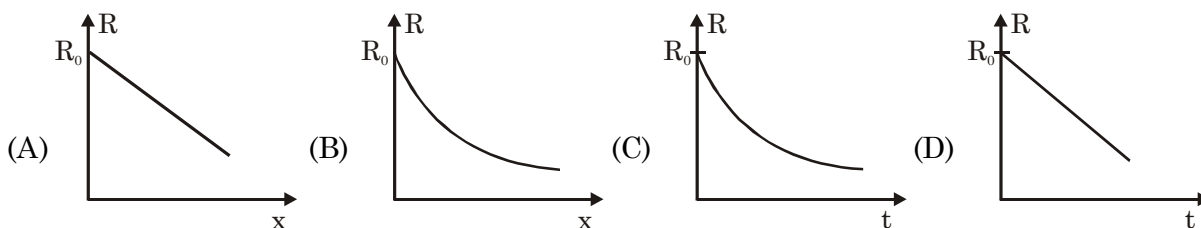
- (A) The induced emf between terminals P and Q is $\frac{\mu_0 N i \omega a^2}{2\ell}$, when current in the coil is i .
- (B) Current in the coil can increase or decrease with time depending upon the value of ω .
- (C) $\omega_{\min}$ for increasing the current in the coil always is $\frac{\ell R}{2\mu_0 N a^2}$
- (D) $\omega_{\min}$ for increasing the current in the coil always is $\frac{2\ell R}{\mu_0 N a^2}$

Space for Rough Work

10. A light string of length 1 m and mass 10 g is tied at one end to a fixed support and other end to a heavy string of length 3m and mass 270 g as shown in the figure. Both the strings are kept at a constant tension of 16 N. The whole system is excited by a source of transverse vibrations of varying frequencies, such that the light string supports one or more loops. Select the correct statement/s from the following :-

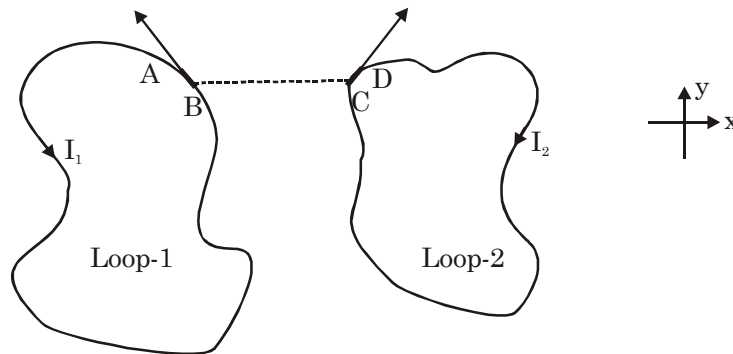


- (A) If the point P corresponds to a node, the minimum number of loops formed along the heavy string will be 9.
 (B) The point P becomes an antinode for the minimum frequency of vibration of 30 Hz.
 (C) The point P is a node, if the possible frequencies of vibration are integral multiple of 10Hz.
 (D) For P to be an antinode, the minimum number of loops formed in the heavy string must be 14.
11. A particle of mass m and charge q is projected into a region having a uniform magnetic field B_0 . Initial velocity (v_0) of the particle is perpendicular to the magnetic field. Apart from the magnetic force the particle faces a frictional force which has a magnitude of $f = kv$ where v is instantaneous speed and k is a positive constant. Which of the following is/are correct. (R is radius of curvature)



Space for Rough Work

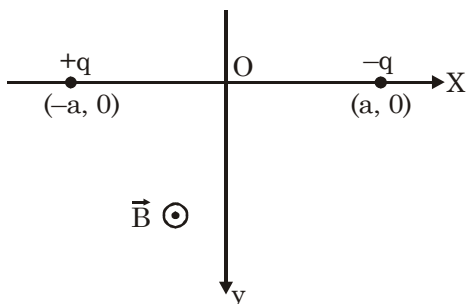
12. Consider two arbitrary elementary current elements AB & CD of lengths dl_1 and dl_2 on the loop-1 and loop-2 respectively. Let $d\vec{F}_{12}$ be the force which the current carrying element $I_2 d\vec{\ell}_2$ exerts on the current carrying element $I_1 d\vec{\ell}_1$ & $\vec{F}_{12}$ represent the net force on 1st loop by the 2nd loop, developed due to interaction between current carrying loops. Similarly $d\vec{F}_{21}$ be the force which the current carrying element $I_1 d\vec{\ell}_1$ exerts on the current carrying element $I_2 d\vec{\ell}_2$ & $\vec{F}_{21}$ represent the net force on 2nd loop by the 1st loop, develop due to interaction between current carrying loops. Choose the **CORRECT** option (s)



- (A) $d\vec{F}_{12} + d\vec{F}_{21} = \vec{0}$ (B) $d\vec{F}_{12} + d\vec{F}_{21} \neq \vec{0}$ (C) $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$ (D) $\vec{F}_{12} + \vec{F}_{21} \neq \vec{0}$
13. A standing wave exists in a string of length 150 cm and is fixed at both ends. The displacement amplitude of a point at a distance of 10 cm from one of the ends is $5\sqrt{3}$ mm. The distance between the two nearest points, within the same loop and having displacement amplitude equal to $5\sqrt{3}$ mm, is 10cm. Which of the following is/are correct :-
- (A) The maximum displacement amplitude in the string is 10 mm.
 (B) The maximum displacement amplitude in the string is 15 mm.
 (C) String is vibrating in a 4th overtone.
 (D) String is vibrating is 5th overtone.

Space for Rough Work

14. Two particles have equal mass m and electric charge of equal magnitude (q) and opposite sign. The particles are held at rest at co-ordinates $(-a, 0)$ and $(a, 0)$ as shown in the figure. The particles are released simultaneously. Consider only the electrostatic force between the particles and the force applied by the external magnetic field on them. Which of the following is/are correct :-



- (A) The speed of negatively charged particle at $x = \frac{a}{2}$ is $\sqrt{\frac{kq^2}{2ma}}$.
- (B) The speed of positively charged particle at $x = -\frac{a}{2}$ is $\sqrt{\frac{kq^2}{ma}}$.
- (C) Y-component of velocity of negative charge particle at $x = \frac{a}{2}$ is $\frac{qBa}{m}$.
- (D) Y-component of velocity of negative charge particle at $x = \frac{a}{2}$ is $\frac{qBa}{2m}$.

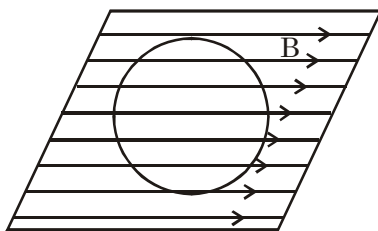
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

A uniform rigid ring of radius r and unknown mass is placed on a non-conducting frictionless horizontal magnetic field of induction B as shown in figure.



15. The minimum current I_0 required to flip over the ring is made to circulate in the coil. Then unknown mass is :-

(A) $\frac{2\pi I_0 r B}{g}$ (B) $\frac{\pi I_0 r B}{g}$ (C) $\frac{\pi I_0 r B}{2g}$ (D) $\frac{3\pi I_0 r B}{2g}$

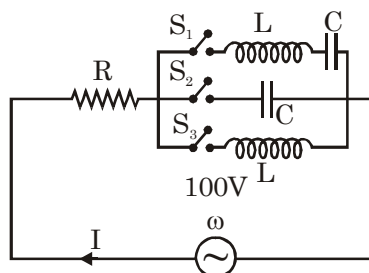
16. A current $I = 2I_0$ is made to circulate in the coil. Find the normal reaction from the table top on the ring immediately after current is switched on :-

(A) $\frac{5\pi r I_0 B}{6}$ (B) $\frac{2}{3}\pi r I_0 B$ (C) $\frac{5\pi r I_0 B}{3}$ (D) $\frac{\pi r I_0 B}{3}$

Space for Rough Work

Paragraph for Questions 17 and 18

In the circuit shown in the figure, one of the three switches is kept closed and other two are open. The value of resistance is $R = 20\Omega$. When the angular frequency (ω) of the 100 V source is adjusted to 500 rad/s, 1000 rad/s and 2000 rad/s it was found that the current I was 4A, 5A and 4A respectively.



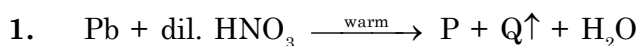
17. Which switch is closed ?
 (A) S_1 (B) S_2 (C) S_3 (D) Data insufficient
18. The value of L and C will be :-
 (A) 5 mH, 200 μF (B) 10 mH, 100 μF (C) 20 mH, 50 μF (D) 25 mH, 40 μF

Space for Rough Work

PART-2 : CHEMISTRY

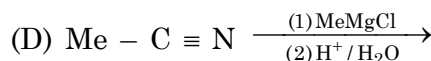
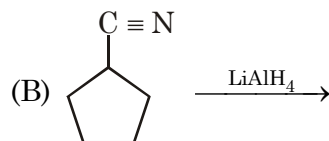
SECTION-I(i) : (Maximum Marks : 18)

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- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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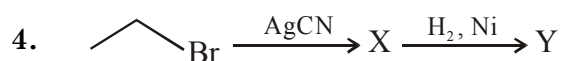


INCORRECT statement for Q is.

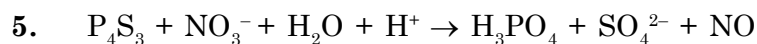
- (A) Paramagnetic colourless gas
 (B) It is oxidized to paramagnetic coloured gas by air
 (C) It combines with $\text{Fe}_2(\text{SO}_4)_3$
 (D) It is also obtained by disproportionation of HNO_2
2. Colour of CrO_4^{2-} (aq.) solution is not changed by :
 (A) dil. HCl (B) NH_3 solution (C) CH_3COOH (D) conc. H_2SO_4
3. In which reaction(s) aldehyde or ketone is not formed as major product?



Space for Rough Work



The final product (Y) is :



Rate of disappearance of P_4S_3 is how many times of rate of disappearance of H_2O in above reaction



6. Mole fraction of methanol in methanol-water solution is 0.4. If partial molar volume of methanol in solution is $40 \text{ cm}^3/\text{mol}$ then find partial molar volume of water (cm^3/mol)
[Given : Density of solution = 0.9 gm/cm^3]



Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

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-
7. Which of the following solutions may show buffer action effectively when equal concentration of two are mixed -
- (A) $\text{H}_2\text{CO}_3 + \text{Na}_2\text{CO}_3$ (B) $\text{Na}_3\text{PO}_4 + \text{H}_3\text{PO}_4$
(C) $\text{CH}_3\text{COOH} + \text{NaOH}$ (D) $\text{HCN} + \text{NaCN}$ (in volume ratio 200 : 1)
8. In which of the following crystal systems unit cell have either only two equal axial distances of unit cell or only two equal axial angles of unit cell or both -
- (A) Tetragonal (B) Orthorhombic (C) Trigonal (D) Monoclinic
9. Solution is prepared by dissolving 1.5 gm of a monoacidic base into 1.5 kg of water at 300 K, which show depression in freezing point by 0.165°C . When 0.496 gm of same base is dissolved in water, require 40 ml of semi molar H_2SO_4 solution for neutralisation.
(Given, $K_{\text{f}(\text{H}_2\text{O})} = 1.86 \text{ K Kg mol}^{-1}$; $M \approx m$, $R = 0.08 \text{ atm-L/mol}$, $\log 2 = 0.30$)
Select **CORRECT** statement for first solution.
- (A) Base is 10% ionised in solution
(B) pH of solution is nearly 11.9
(C) Osmotic pressure of solution is 2.13 pascal
(D) Observed molar mass of base in solution is less than 12.4

Space for Rough Work

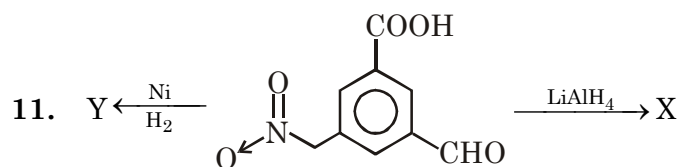
10. Which of the following decay may increase $\left(\frac{n}{p}\right)$ ratio -

(A) α

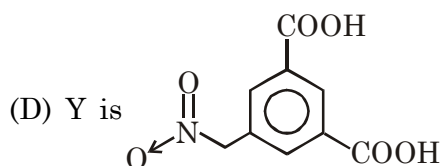
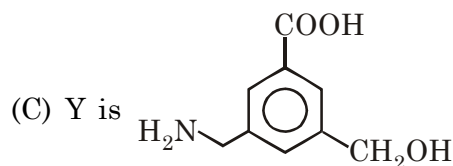
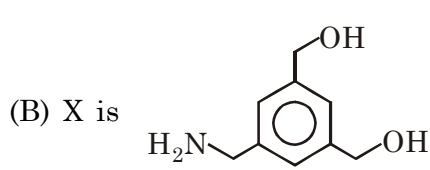
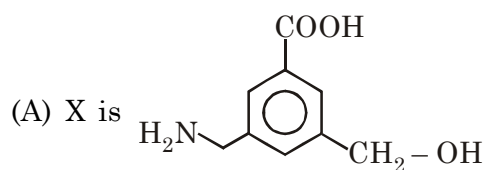
(B) β

(C) positron

(D) K-capture

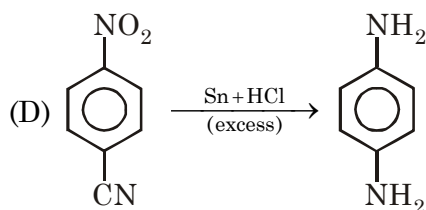
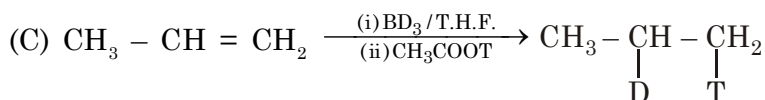
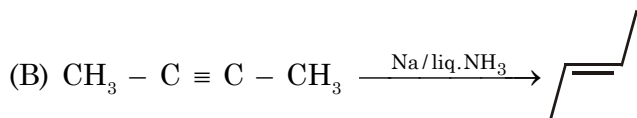
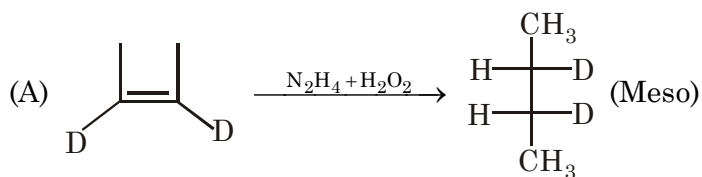


Select correct option(s)?



Space for Rough Work

12. Identify correct reaction sequence?



13. Identify compounds give positive $\text{Br}_2 + \text{H}_2\text{O}$ test (Decolourise $\text{Br}_2 + \text{H}_2\text{O}$)



14. Identify total number of ketones with molecular formula $\text{C}_6\text{H}_{12}\text{O}$ give racemic mixture of alcohol on reaction with LiAlH_4 followed by H_2O ?



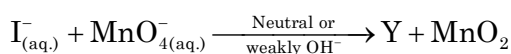
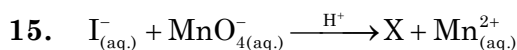
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

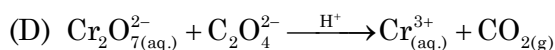
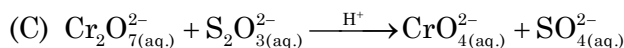
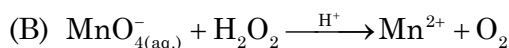
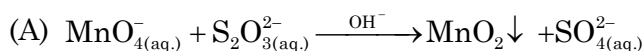
KMnO_4 and $\text{K}_2\text{Cr}_2\text{O}_7$ are important industrial reagents used for oxidising purpose. Oxidising behaviour of both reagents changes with change in pH.



Products X and Y are respectively.

- (A) I_3^- , I_2 (B) I_2 , IO_3^- (C) I_2 , I_3^- (D) IO_3^- , I_2

16. Which of the following reaction is **INCORRECT**.



Space for Rough Work

Paragraph for Questions 17 and 18

A complex compound of cobalt contain five H_2O molecule, one nitro group and two bromide ion for one Co^{3+} cation. One molecule of this compound produce three ions in aqueous solution, on reacting with excess of AgNO_3 solution, two moles of AgBr get precipitated.

17. The type of isomerism shown by the complex compound is

- (A) Geometrical , Ionization
- (B) Ionization, Linkage
- (C) Linkage, Optical
- (D) Geometrical, Optical

18. Magnetic moment of complex compound is.

- (A) 0 B.M. (B) $\sqrt{24}$ B.M. (C) $\sqrt{15}$ B.M. (D) $\sqrt{3}$ B.M.

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 18)**

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $f(x) = \frac{x}{1+e^{\frac{1}{x}}}$, $x \neq 0$, then-

(A) $x^3 f'(x) - x^2 f(x) - e^{\frac{1}{x}} (f(x))^2 = 0$

(B) $x^3 f'(x) + x^2 f(x) - e^{\frac{1}{x}} (f(x))^2 = 0$

(C) $xf'(x) - f(x) + e^{\frac{1}{x}} (f(x))^2 = 0$

(D) $xf'(x) - f(x) - e^{\frac{1}{x}} (f(x))^2 = 0$

2. The length of edges of rectangular parallelepiped meeting in a point P are α, β, γ & a plane is drawn cutting these edges at a distance α', β', γ' ($\alpha', \beta', \gamma' \neq 0$) respectively from P, meets the

body diagonal PQ of parallelepiped at Q', then $\frac{PQ}{PQ'}$ is -

(A) $\frac{\alpha + \beta + \gamma}{\alpha' + \beta' + \gamma'}$

(B) $\frac{\alpha \cdot \beta \cdot \gamma}{\alpha' \beta' \gamma'}$

(C) $\left(\frac{\alpha \cdot \beta \cdot \gamma}{\alpha' \beta' \gamma'} \right)^2$

(D) $\frac{\alpha}{\alpha'} + \frac{\beta}{\beta'} + \frac{\gamma}{\gamma'}$

Space for Rough Work

3. Let $\vec{p}, \vec{q}, \vec{r}$ be three unit vectors such that $\vec{p} \times \vec{q} = \vec{r}$. If $\vec{a}$ is any vector such that $[\vec{a} \vec{q} \vec{r}] = 1, [\vec{a} \vec{r} \vec{p}] = 2$ and $[\vec{a} \vec{p} \vec{q}] = 3$ then $\vec{a} =$
- (A) $\vec{p} + 3\vec{q} + \vec{r}$ (B) $3\vec{p} + 2\vec{q} + \vec{r}$ (C) $\vec{p} - 2\vec{q} - 3\vec{r}$ (D) $\vec{p} + 2\vec{q} + 3\vec{r}$
4. If area bounded by $y = x^2 - 2x - 1$ and $mx + y - 1 = 0$ is minimum, then m is equal to
- (A) 1 (B) 2 (C) 3 (D) 4
5. If $\int \frac{\operatorname{cosec}^2 x}{(\operatorname{cosec} x + \cot x)^{9/2}} dx = (\operatorname{cosec} x - \cot x)^{7/2} \left(\frac{1}{\alpha} + \frac{(\operatorname{cosec} x - \cot x)^2}{11} \right) + C$
(where C is constant of integration and $\alpha \in \mathbb{N}$), then α is -
- (A) 5 (B) 9 (C) 10 (D) 7
6. Let M be a 3×3 matrix satisfying $M^3 = O$. Then value of $\det\left(\frac{1}{2}M^2 + M + I\right)$ can not be
- (A) 0 (B) 1 (C) 2 (D) 3
- (where I is unit matrix of order 3, O is null matrix of order 3)

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Given a differentiable function $f(x)$ satisfying the integral equation

$$f(x) = \int_0^x e^{x-y} f'(y) dy - (x^2 - x + 1)e^x \forall x \in \mathbb{R}, \text{ then-}$$

- (A) $f(x)$ is local minimum at $x = -\frac{1}{2}$
- (B) The equation $f(x) + \log_2 x = 0$, has exactly one real solution
- (C) If $f(x)$ has an extremum point at $x = x_1$ and point of inflection at $x = x_2$, then $|x_1 - x_2| = 1$
- (D) $|f(x) - f'(x)| = |f'(x) - f''(x)|$

Space for Rough Work

8. Let $\vec{v}_1, \vec{v}_2, \vec{v}_3$ be three vectors such that $|\vec{v}_1|^2 = |\vec{v}_2|^2 = |\vec{v}_3|^2 = 3$ and

$$|\vec{v}_1 + \vec{v}_2 - \vec{v}_3|^2 + |\vec{v}_2 + \vec{v}_3 - \vec{v}_1|^2 + |\vec{v}_3 + \vec{v}_1 - \vec{v}_2|^2 = 36, \text{ then}$$

(A) $\vec{v}_1 \cdot \vec{v}_2 + \vec{v}_2 \cdot \vec{v}_3 + \vec{v}_3 \cdot \vec{v}_1 = \frac{-9}{2}$

(B) $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are coplanar vectors

(C) $\vec{v}_1 \times \vec{v}_2 = \vec{v}_2 \times \vec{v}_3 = \vec{v}_3 \times \vec{v}_1 \neq \vec{0}$

(D) $\vec{v}_1, \vec{v}_2, \vec{v}_3$ represent the sides of triangle.

9. Let $f(x) = \begin{cases} (x-1)(x-n) & x \leq n \\ x^2 - n & x > n \end{cases}$, then identify correct statement(s) ($n \in \mathbb{I}$)

(A) $\forall n \leq 0$ $f(x)$ has local minima at $x = n$

(B) $\forall n \geq 0$ $f(x)$ has local minima at $x = n$

(C) $\forall n > 0$ $f(x)$ is strictly increasing at $x = n$

(D) $f(x)$ has always a local minima $\forall n \in \mathbb{I}$

10. If position vectors of vertices A, B, C and D of tetrahedron with respect to its centroid are $\hat{i} + 2\hat{j} + 2\hat{k}$, $2\hat{i} + 3\hat{j} - 4\hat{k}$, $-4\hat{i} - 6\hat{j} + 4\hat{k}$ and $\hat{i} + \hat{j} - 2\hat{k}$ respectively, then-

(A) Volume of tetrahedron is $\frac{8}{3}$ cu. units

(B) $AD > AB$

(C) Area of $\triangle ABD$ is $\frac{\sqrt{117}}{2}$ sq. units

(D) $\triangle DAB$ is an equilateral triangle

Space for Rough Work

11. Let $f(x) = e^x \sqrt{x}$ ($x > 0$) & $g(x)$ be its inverse function. If $I = \int_0^1 f(x) dx$, then-

(A) $\int_0^e g(x) dx = e - I$

(B) $\int_0^1 \frac{f(x)}{x} dx = 2(e - I)$

(C) $I > 1$

(D) $I < 1$

12. Consider three square matrix A, B, C of order 3 such that $A^T = A - B$, $B^T = B - C$, then choose correct option(s) (Here X^T denotes transpose of matrix X)

(A) $|C| = 4|B|$

(B) $|B| = 2|C|$

(C) $|A| = 0$

(D) $C = 2B$

13. Let matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$, then-

(A) $(A + B)^{2005} = A^{2005} + B^{2005}$

(B) $(A - B)^{2005} = A^{2005} - B^{2005}$

(C) $(3A + 7B)^n = 3^{n-1} (3^n A + 7^n B)$, $n \in \mathbb{N}$

(D) $(3A + 7B)^n = 3^{2n} A + 3^n 7^n B$, $n \in \mathbb{N}$

14. For curve $f(x) = 1 + \sqrt{1 - x^2}$.

(A) Equation of tangent of finite slope from point (1, 3) on $y = f(x)$ is $3x - 4y + 9 = 0$

(B) Number of tangents to $y = f(x)$ drawn from origin is one.

(C) Equation of chord of curve with mid point $P\left(\frac{1}{2}, \frac{3}{2}\right)$ is $x + y = 2$

(D) Area bounded by $y = f(x)$ and $|y| = |x|$ is $\frac{\pi}{2} + 1$ square units.

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

ABCD is a regular tetrahedron. M(0, 0, 0) is the midpoint of the altitude DN of the tetrahedron. Coordinates of the points A and B are (0, 0, -2) and (b, b, 0) respectively (with $b > 0$). Also x-coordinate of the point C is given to be positive

15. Image of the point D about the plane generated by ABC is
- (A) $(2\sqrt{2}, 0, -2)$ (B) $(2, 0, -2)$ (C) $(2, 0, -2\sqrt{2})$ (D) None of these
16. Volume of the tetrahedron ACBM is
- (A) $\frac{5\sqrt{2}}{6}$ (B) $\frac{4\sqrt{2}}{3}$ (C) $\frac{4}{3}$ (D) None of these

Space for Rough Work

Paragraph for Questions 17 and 18

If $f(x,y) = 0$ be the solution of differential equation

$$(2y \operatorname{cosec} 2x + \ln \cot y) dx + (\ln \tan x - 2x \operatorname{cosec} 2y) dy = 0 \text{ such that } f\left(\frac{\pi}{4}, \frac{\pi}{2}\right) = 0, \text{ then}$$

17. $f(x,y)$ is-

(A) $(\tan x)^x (\cot y)^y$

(B) $(\tan x)^y (\cot y)^x$

(C) $(\tan x)^{x^2} (\cot y)^{y^2}$

(D) $(\tan x)^{y^2} (\cot y)^{x^2}$

18. If $\int_0^{\infty} \frac{f\left(\frac{5\pi}{4}, \frac{2017\pi}{4}\right) \cos x}{x} dx = \frac{\pi}{2}$, then $\int_0^{\infty} \frac{\cos^3 x}{x} dx$ is equal to-

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{2}$

(D) π

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has one section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	8	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	32
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator